

Valve After-Sales Guidance Manual

During the use of valves, various types of failures can occur. Generally, there are two main factors: first, failures are related to the number of components that make up the valve - more parts means more potential failures; second, failures are closely related to the valve's design, manufacturing, installation, operating conditions, operation, and maintenance quality. When each aspect is properly handled, valve failures can be greatly reduced.

Section 1: Prevention and Treatment of Common Failures of General Valve Parts

Valves typically consist of components such as valve body, valve cover, packing, gaskets, sealing surfaces, valve stem, bracket, and transmission devices, collectively referred to as general valve parts. Below is a detailed introduction to common valve failures, their causes, preventive measures, and troubleshooting methods.

I. Common Failures of Valve Body and Valve Cover

Below is detailed information on common failures, preventive measures, and troubleshooting methods for valve bodies and covers.

(1) Common Failures: Damage and Leakage

Causes of Failure:

- ① Poor design (insufficient safety factor, unreasonable structure, excessive stress concentration).
- ② Poor forging and casting quality (folds, cold shuts, pores, slag inclusions, loose structures, hidden cracks, uneven thickness, uneven material, material not meeting design requirements, not manufactured according to drawings).
- ③ Poor welding (welding defects, overly brittle welds, excessive stress causing cracks).
- ④ Incorrect installation (misalignment and distortion).
- ⑤ Improper selection (valve unsuitable for operating conditions causing rupture).
- ⑥ Valve internal pressure and temperature too high with significant fluctuations.
- ⑦ Water hammer damaging the valve.
- ⑧ Freezing damage.
- ⑨ Accidental impact.
- ⑩ Fatigue damage.

Preventive Measures:

- ① Design should comply with national standards and relevant specifications, structure should be reasonable, avoiding excessive stress concentration. New products should be field-tested before mass production.

- ② Strictly follow operating procedures and process discipline, forge and process according to drawings and technical requirements; establish a complete quality assurance system, important valves should undergo material analysis and flaw detection, strength tests should be conducted before delivery, testers and assemblers should sign for responsibility.
- ③ Strictly follow welding procedures, conduct thorough inspection and flaw detection after welding, strength tests should be conducted before delivery, testers should sign for responsibility.
- ④ Install centered, even force distribution, prevent flange misalignment and gaps. Large valves should have support brackets. Cast iron and non-metal valves are brittle and require special attention.
- ⑤ Strictly select valves according to operating conditions (medium, temperature, pressure), important areas and valves with poor operating conditions should have sufficient margin.
- ⑥ Safety systems should be sensitive, pressure and temperature displays should be correct, operations should be smooth, emergency measures should be effective.
- ⑦ Operations should be smooth, have water hammer prevention devices, prevent sudden pump stoppage and fast valve closing.
- ⑧ Valves should have heat tracing or heating facilities, valves not in use in cold weather should have medium removed.
- ⑨ Do not stack heavy objects, prevent impact on valves during construction.
- ⑩ Valves that exceed service life or show early fatigue defects should be replaced.

Troubleshooting Methods:

- ① Stop machine, relieve pressure, repair or replace valve.
- ② Repair damaged valve using mechanical plugging, welding plugging, adhesive plugging, or pressurized glue methods.
- ③ Replace valve damaged parts under pressure or replace entire valve.

(2) Common Failures: Sand Hole Leakage**Causes of Failure:**

- ① Casting defects (pores, slag inclusions, loose structures).
- ② Forging defects (slag inclusions, folds).
- ③ Injection molding defects (pores, slag inclusions, cold shuts, missing material).
- ④ Welding defects (pores, slag inclusions, undercut, incomplete penetration).

Preventive Measures:

Strictly follow operating procedures and process discipline, establish complete quality assurance system, strength tests should be conducted before delivery, testers should sign for responsibility.

Troubleshooting Methods:

- ① Stop machine, relieve pressure, repair or replace valve.
- ② Repair using mechanical plugging, welding plugging, adhesive plugging, pressurized glue, physical plugging, or chemical plugging methods.

(3) Common Failures: Aging Leakage**Causes of Failure:**

- ① Poor manufacturing quality (plastic and rubber have poor corrosion resistance, not resistant to aging).

- ② Improper selection (does not meet operating conditions).
- ③ Ineffective anti-aging measures.
- ④ Poor maintenance (replacement not timely).

Preventive Measures:

- ① Strictly follow process and operating discipline, add anti-aging agents on time, in correct amounts, according to quality requirements. Have complete quality assurance system.
- ② Select valves according to operating conditions, pay attention to constraints among pressure, temperature, and medium, leave adequate margin.
- ③ Based on different properties of plastic and rubber valves, properly perform heat, cold, sun, and dust protection.
- ④ Maintain valves according to schedule, promptly replace valves showing aging signs.

Troubleshooting Methods:

- ① Stop machine, relieve pressure, repair or replace valve.
- ② Replace valve under pressure using seal replacement method or freezing method.
- ③ For locally aged damaged valves, use relevant methods for on-pressure repairs.

(4) Common Failures: Corrosion Leakage

Causes of Failure:

- ① Poor manufacturing quality (slag inclusions, uneven structure, or material not meeting design requirements).
- ② Welding defects (slag inclusions, structure burn, or weld metal not matching base metal, not resistant to medium).
- ③ Improper selection (not resistant to medium corrosion).
- ④ Ineffective corrosion protection.
- ⑤ Poor maintenance (replacement not timely).

Preventive Measures:

- ① Strictly follow process and operating discipline, have complete quality assurance system, control factory quality.
- ② Strictly follow welding procedures, have quality assurance measures.
- ③ Select valves according to medium corrosion performance and operating conditions.
- ④ Adopt various anti-corrosion measures (coating, plating, electrode protection, corrosion inhibitors).
- ⑤ Maintain according to schedule, promptly replace corroded valves.

Troubleshooting Methods:

- ① Stop machine, relieve pressure, repair or replace valve.
- ② Repair using mechanical plugging, welding plugging, adhesive plugging, pressurized glue, chemical plugging, or physical plugging methods.

II. Common Packing Failures

Below is detailed information on common packing failures, preventive measures, and troubleshooting methods.

(1) Common Packing Failure: Insufficient Preload

Causes of Failure:

- ① Too little packing (too little packing during installation, or preload reduced due to wear, aging, and improper assembly).
- ② No preload clearance.
- ③ Packing compression ring stuck (compression ring tilted or diameter too large).
- ④ Thread obstruction (crossed threads, rust, foreign matter).

Preventive Measures:

- ① Install sufficient packing as specified, timely replace expired packing, correctly assemble packing, prevent top-tight bottom-loose defects.
- ② Compression ring should be inserted to depth of 1/4 to 1/3 of its height.
- ③ Before installing packing, check clearance, prevent compression ring tilt, prevent packing exposure.
- ④ Regularly inspect and clean bolts, apply graphite powder when tightening.

Troubleshooting Methods:

- ① Close valve or engage upper seal, repair parts, add packing, adjust preload.
- ② Use expansion method or pressurized glue method for plugging.
- ③ Check reason for compression ring stuck, file to normal size if needed.
- ④ For thread obstruction, clean with rust penetrator or kerosene, file thread to proper tightness.